

Kestrel Trading System

Signal Pipeline + Trade Execution

Powered by Kestrel x Freqtrade

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All data simulated - dry run mode

1. System Overview

Kestrel is an automated signal ingestion and trading pipeline. It collects data from 400+ sources (Telegram channels, RSS feeds, Substack newsletters, on-chain data), scores every event for trading relevance, and feeds high-confidence signals into Freqtrade for paper trade execution on Coinbase Advanced.

Everything runs on a local Linux server with zero cloud dependency. Data lands in DuckDB (local database), signals are scored every 5 minutes, and trades are simulated with a \$1,000 paper wallet.

Architecture Flow

SOURCES --> DuckDB --> Scorer --> Freqtrade --> Paper Trades

Sources include:

- Whale Alert - on-chain whale transactions (lead indicator, score 0.38)
- DefiLlama - TVL and DeFi flow data (catalyst, score 0.34)
- Cointelegraph - crypto news feed (30 articles/cycle)
- Disclose.tv - geopolitical events
- TLDR Newsletter - tech news summary
- Bankless, Hacker News, TechCrunch, a16z Crypto, Coindesk
- Total: 418 distinct sources, 18 event types

Timeline - What Was Built (June 7, 2025)

- Pipeline: Full ingestion pipeline from 400+ sources into DuckDB
- Scorer: 9-step post-ingest scoring engine with time-decay and source priors
- Ranking: Cross-source agreement buckets, top-20 ranked queue
- Freqtrade: Live on Coinbase Advanced, dry-run \$1000, BTC/ETH/SOL
- Strategy: Basis divergence (0.3%) + DuckDB signal confirmation
- Watchdog: Freshness monitor every 60s, auto-quarantine stale sources
- Budget: \$50/day cap with \$5 floor, \$10 per-trade limit

2. Signal Pipeline - How It Works

Step 1: Ingestion

Sources are collected via Telegram channels (HTML parsing), RSS feeds, and Substack newsletters. Each event is normalized into a standard format: source_id, event_type, timestamp, headline, and raw payload. Data lands in the DuckDB events table.

Step 2: Scoring (Every 5 Minutes)

The scorer loads new unscored events and computes an edge_score:

```
edge_score = recency_weight x novelty x asset_relevance
```

```
    x source_prior x tier_mult x cross_source_boost
```

```
    x (1 - false_positive_penalty)
```

- recency_weight: Time-decay based on tier half-life (lead=30min, catalyst=2hr, narrative=12hr)
- source_prior: Pre-calibrated confidence per source (Whale Alert=0.90, Cointelegraph=0.66)
- tier_mult: Lead indicators get 1.15x, narrative gets 0.90x, archival gets 0.50x
- cross_source_boost: If 3+ sources agree on an event, score gets boosted
- false_positive_penalty: Historical accuracy penalty per source

Step 3: Ranking

Top 20 signals by edge_score get written to pulse/ranked-queue.json. The queue updates every 5 minutes and scores decay naturally.

Step 4: Trade Execution

Freqtrade (StrikerBasisStrategy) reads the ranked queue from DuckDB and enters trades when:

- Basis divergence > 0.3% (perp vs spot spread)
- RSI < 70 (not overbought for longs)
- Pipeline signal confidence >= 0.15
- Daily budget remaining >= \$5.00

3. DuckDB Database

All data is stored locally in /home/synczus/kestrel/signals.duckdb (23 MB).

Table: signal_scores (4,671 rows)

- score_id - Unique identifier
- source_id - Source name (whale-alert, defillama, etc.)
- asset_symbol - BTC, ETH, SOL (or null for general news)
- event_type - general_news, whale_transfer, geopolitical_event, etc.
- edge_score - Composite score 0.0-1.0 (highest: 0.38)
- tier - lead_indicator, catalyst_confirmation, narrative_formation, archival_reference
- event_ts - When the event happened
- scored_at - When it was scored
- 22 columns total, including raw_payload_json

Table: trade_log (0 trades - dry run mode)

- trade_id - Unique trade identifier
- pair - BTC/USDC, ETH/USDC, SOL/USDC
- side - long
- amount - Trade quantity
- price - Execution price
- amount_usd - Dollar amount (max \$10/trade)
- signal_score - Edge score that triggered the trade
- source - Which source triggered it
- engine - freqtrade
- executed_at - Timestamp

Top 5 Current Signals

1. whale-alert: general_news score=0.384 tier=lead_indicator

2. defillama: defi_tvl_flow score=0.340 tier=catalyst_confirmation

Ranked queue updates every 5 minutes with natural score decay

Note: 0 trades in trade_log because Freqtrade is in dry-run mode. Simulated trades are recorded in Freqtrade's own SQLite database instead.

4. Freqtrade Configuration

Exchange Settings

Exchange	Coinbase Advanced Trade
Trading Pairs	BTC/USDC, ETH/USDC, SOL/USDC
Trading Mode	Spot (no leverage)
Run Mode	Dry Run (paper trading)
Timeframe	5-minute candles

Wallet Settings

Paper Balance	\$1,000 USDC
Max Open Trades	3
Stake per Trade	\$10 USDC
Daily Cap	\$50 (with \$5 floor)
Tradable Balance	95% of wallet

Strategy: StrikerBasisStrategy

- Primary signal: Basis divergence between perpetual and spot markets
- Confirmation: DuckDB pipeline scores (threshold: 0.15)
- RSI filter: Buy when RSI < 70, sell reversal when RSI reversal
- Custom position sizing: Score-scaled from \$3 to \$10 per trade

Risk Management

- Budget guard: \$50/day maximum total spend recorded in DuckDB
- Per-trade cap: \$10 maximum per entry
- Trading stops when budget remaining < \$5
- Trailing stop: 2% initial, tightens to 0.5% after 1% profit
- ROI targets: 1% immediate, 0.5% after 1hr, 0.1% after 2hrs
- Force entry enabled for bridge-driven trades

API Status

REST API	http://127.0.0.1:8081
Status	Online - heartbeat confirmed
Auth	HTTP Basic Auth configured

5. How a Signal Becomes a Trade

Walk through of one complete cycle from detection to trade:

Example: Whale Alert detects a large BTC transfer

1. Telegram message arrives: '2,500 BTC (\$165M) transferred from unknown wallet to Binance'
2. Pipeline ingests and normalizes the event:
 - source_id = whale-alert
 - event_type = whale_transfer
 - asset = BTC
 - timestamp = now
3. Scorer runs (every 5 min):
 - Recency weight: high (fresh event, within 30-min lead-indicator half-life)
 - Source prior: 0.90 (Whale Alert is our highest-trust source)
 - Tier: lead_indicator (highest multiplier: 1.15x)
 - Cross-source boost: moderate if other sources also covering
 - Final edge_score: ~0.35 to 0.45
4. Ranked queue updates with Whale Alert signal at #1
5. On next 5-minute candle, StrikerBasisStrategy reads DuckDB:
 - Checks budget: \$50/day available
 - Checks basis divergence: Is perp > spot? (confirms bullish pressure)
 - Checks RSI: < 70 (not overbought)
 - Signal confidence 0.38 passes threshold of 0.15
6. Trade executes (simulated):
 - Entry: Buy BTC/USDC at market price
 - Amount: ~\$10 (score-scaled position)
 - Stoploss: 2% trailing
 - Take profit triggered by RSI reversal or basis convergence
7. Trade is journaled:
 - Freqtrade records in local SQLite
 - Bridge writes to DuckDB trade_log (when active)
 - Daily budget updated, guard recalculates

6. Roadmap & Next Steps

Immediate Next Steps

- Bridge Deployment: Wire `striker_bridge.py` for automatic `force_entry` from DuckDB queue
- Supervisor: Process manager that launches `Freqtrade` + `bridge` + `budget guard`
- Systemd Timers: `Scorer` every 60s, `watchdog` every 60s, `compound-state` every 5min
- DuckDB Lock Fix: Serialize concurrent writers (flock)

Live Trading Path

- Flip `dry_run = false` in `config.json`
- Add Coinbase Advanced API keys
- Fund wallet with \$50+ USDC
- Supervisor auto-verifies 3+ clean paper trades first

Coming Next

- CryptoQuant API: On-chain exchange flow data (highest signal density)
- a16z Crypto Substack: Structural narratives (RSS verified working)
- TechCrunch + Coinstack + The Tech Buzz: Additional RSS feeds

Provara Integration

- Every trade decision gets Ed25519-signed into a Provara vault
- Verifiable audit trail for why every trade was entered/exited
- Provara = trust layer, DuckDB = analytics layer - complementary
- Dashboards on DuckDB data, cryptographic proofs in Provara

7. Summary Statistics

Total signals scored	4,671
Active source IDs	418
Event type categories	18
Highest edge_score	0.384 (Whale Alert)
Top scoring tier	lead_indicator
Database size	23 MB (DuckDB)
Freqtrade pairs	BTC/USDC, ETH/USDC, SOL/USDC
Paper wallet balance	\$1,000 USDC
Daily budget cap	\$50/day
Per-trade limit	\$10
Scorer runs every	5 minutes
Watchdog checks every	60 seconds
API server	Port 8081, online

This report reflects the current state of the Kestrel trading system as of the generation date. All trades are simulated in dry-run mode. No real funds have been deployed. The system is designed for educational and research purposes.